QIP 1 – Riassunto

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1 No-cloning theorem

Suppose we want to copy a state $|\psi\rangle$: we want an operation that maps

$$|\psi\rangle \otimes |0\rangle \longmapsto |\psi\rangle \otimes |\psi\rangle = \alpha^2 |00\rangle + \beta^2 |11\rangle + \alpha\beta |01\rangle + \alpha\beta |10\rangle .$$

A CNOT works for $|\psi\rangle = |0\rangle, |1\rangle$ (similarly, the CNOT with control and target swapped, with the second qubit prepared in $|+\rangle$, copies $|+\rangle, |-\rangle$), but doesn't work for a general $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$:

$$\mathrm{CNOT}\big(|\psi\rangle \otimes |0\rangle\big) = \alpha |00\rangle + \beta |11\rangle \neq |\psi\rangle \otimes |\psi\rangle .$$

- For a set S of quantum states, suppose there exists a unitary operation U which maps $|\psi\rangle \otimes |0\rangle \mapsto |\psi\rangle \otimes |\psi\rangle \ \forall\, |\psi\rangle \in S$
  $\Rightarrow$ the states in S are mutually orthogonal.

2 Deutsch algorithm

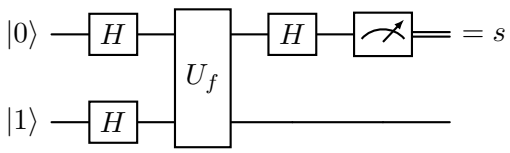
Consider a function $f : \{0,1\} \to \{0,1\}$. There are 4 such functions, 2 of which give constant output.

- Given a black-box implementation of f, decide if f is constant.
  - $\to$ Classical: 2 queries
  - $\to$ Quantum: 1 query

We have U_f: $U_f\big(|x\rangle \otimes |y\rangle\big) = |x\rangle \otimes |f(x) \oplus y\rangle$

- $U_f\big(|x\rangle \otimes |-\rangle\big) = (-1)^{f(x)} |x\rangle \otimes |-\rangle$
- $U_f\big(|+\rangle \otimes |-\rangle\big) = (-1)^{f(0)} \frac{1}{\sqrt{2}}\big(|0\rangle + (-1)^{f(0)\oplus f(1)} |1\rangle\big) \otimes |-\rangle$

$\Rightarrow$ If f constant, we have $|+\rangle |-\rangle$. If f balanced, $|-\rangle |-\rangle$.
We can use H to rotate back to the computational basis:



$$s = \begin{cases} 0 \to \text{constant} \\ 1 \to \text{balanced} \end{cases}$$

3 Postulates of QM

1. To each d.o.f. (quantum system) is associated a Hilbert space $\mathcal{H} \simeq \mathbb{C}^d$ with inner product

$$\langle u|v\rangle = \sum_{j=1}^{d} u_j^* v_j \qquad \forall u, v \in \mathcal{H}.$$

2. Physical observables are Hermitian operators on $\mathcal{H}$.

3. The state of a quantum system is given by a normalized vector in $\mathcal{H}$.

4. A measurement of $\hat{A}$ on system S results in a single value of the set of the eigenvalues of $\hat{A}$. Let Π_a be the projection operator onto the subspace of $|\phi\rangle$: $A|\phi\rangle = a|\phi\rangle$

$$\Rightarrow \text{Born rule:} \quad P(a) = \langle\psi|\Pi_a|\psi\rangle\,, \qquad \langle A\rangle = \langle\psi|A|\psi\rangle\,.$$

Projection postulate:

$$|\psi'\rangle = \frac{\Pi_a|\psi\rangle}{\sqrt{P_a}}.$$

5. Dynamics of quantum systems are described by the Schrödinger equation

$$|\psi_t\rangle = U_t|\psi_0\rangle \qquad \text{for unitary } U.$$

6. Two quantum systems can be regarded as a single system $\mathcal{H}_{AB} := \mathcal{H}_A \otimes \mathcal{H}_B$.

4 Tensor product

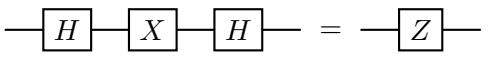
$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},\ B \quad \Rightarrow \quad A \otimes B = \begin{bmatrix} aB & bB \\ cB & dB \end{bmatrix}$$

For column vectors $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$:

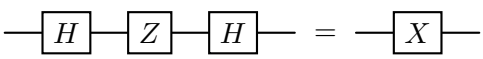
$$|\psi\rangle \otimes |\psi\rangle = \begin{bmatrix} \alpha^2 \\ \alpha\beta \\ \beta\alpha \\ \beta^2 \end{bmatrix}$$

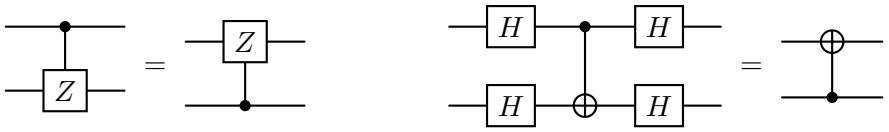
- $A \otimes (B + C) = A \otimes B + A \otimes C$
- $(A \otimes B)(C \otimes D) = AC \otimes BD$

5 Useful circuit identities



$\boxed{S^2 = Z}$





6 Entanglement

We say $|\psi\rangle$ is a PRODUCT STATE if $\exists\ |\varphi\rangle, |\theta\rangle$ s.t. $|\psi\rangle = |\varphi\rangle \otimes |\theta\rangle$.
Any state which is not a product state is an ENTANGLED STATE.

We can use the Schmidt decomposition to tell if a state is entangled:

$$|\psi\rangle_{AB} = \sum_\ell s_\ell \underbrace{|\eta_\ell\rangle \otimes |\xi_\ell\rangle}_{\text{o.n. bases}}$$

Product states have only one $s_\ell \neq 0$.

How to do it:

$$|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle \;\rightsquigarrow\; M = \begin{bmatrix}\alpha & \beta\\ \gamma & \delta\end{bmatrix}$$

- If $\det M \neq 0$, entangled.
- Explicitly: calculate the eigenvalues λ_ℓ of $MM^\dagger \to s_\ell = \sqrt{\lambda_\ell}$.
  It's entangled if both are nonzero.

7 Bell's theorem and CHSH

No local, realistic model can reproduce all of the predictions of QM.

CHSH: consider observables Q, R on system A; S, T on system B, with possible values ± 1.

$$C := QS - QT + RS + RT = (Q+R)S + (R-Q)T = \pm 2 \quad \Rightarrow \quad \langle C\rangle \leq 2$$

QM violates CHSH. Using $|\Phi\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$,

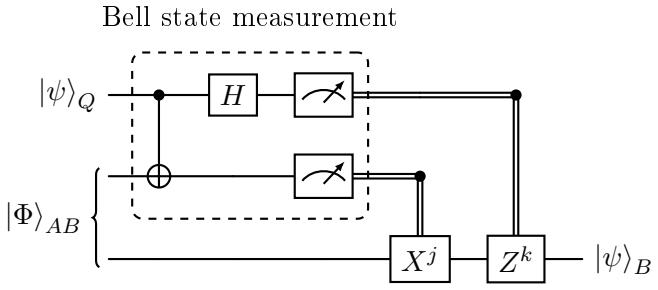
$$Q = \sigma_z, \quad R = \sigma_x, \quad S = \tfrac{1}{\sqrt{2}}(\sigma_x + \sigma_z), \quad T = \tfrac{1}{\sqrt{2}}(\sigma_x - \sigma_z),$$

we get $\langle C\rangle = 2\sqrt{2} \not\leq 2$.

8 Teleportation

Bell basis: $|\Phi_{jk}\rangle = \mathbb{I} \otimes \sigma_x^j \sigma_z^k\,|\Phi\rangle$, with $|\Phi\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)$.

Input: $|\Phi\rangle_{AB}$, arbitrary $|\psi\rangle_Q$. **Output**: $|\psi\rangle_B$.



1. Bell measurement of $QA \to \{j,k\}$
2. Transmit $\{j,k\}$ to Bob
3. Apply $\sigma_z^k \sigma_x^j$ to B

$$\begin{aligned}|\psi\rangle_Q\,|\Phi\rangle_{AB} &= \big(\alpha\,|0\rangle + \beta\,|1\rangle\big) \otimes \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)\\ &= \tfrac{1}{\sqrt{2}}\big(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|100\rangle + \beta\,|111\rangle\big)\\ &\overset{\substack{\text{writing } |00\rangle,|11\rangle,\ldots\\ \text{in Bell basis}}}{=} \frac{1}{2}\sum_{j,k=0}^{1} |\Phi_{jk}\rangle_{QA} \underbrace{\sigma_x^j \sigma_z^k\,|\psi\rangle_B}_{\text{applying } \sigma_x^j\sigma_z^k \text{ undoes}}\end{aligned}$$

Bell basis

$$|\Phi_{00}\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big) \qquad |\Phi_{10}\rangle = \tfrac{1}{\sqrt{2}}\big(|01\rangle + |10\rangle\big)$$
$$|\Phi_{01}\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle - |11\rangle\big) \qquad |\Phi_{11}\rangle = \tfrac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big)$$

9 Classical complexity

Every Boolean function can be computed by a Boolean circuit.

- SIZE: # of gates
- WIDTH: number of bits needed, including scratch
- DEPTH: number of layers needed

An algorithm is considered efficient if it runs in time polynomial in the instance size n ($O(n^c)$ or poly(n)).

P: problems solved in poly(n) on a classical computer

NP: decision problems whose solution can be verified in polynomial time

BPP: problems which can be solved in poly(n) with failure probability $\leq \frac{1}{3}$

P $\subset$ NP, P $\subset$ BPP

10 Quantum complexity

A quantum circuit is composed of 3 steps:

1. Prepare $|0\rangle^{\otimes k}$
2. Apply k-qubit unitary U
3. Measure σ_z on each qubit

Circuit size = time complexity.

BQP: $\exists$ poly-size quantum circuit such that prob(correct) $> \frac{2}{3}$

BPP $\subset$ BQP: quantum is at least as good as classical.

Solovay–Kitaev $\rightarrow$ can approximate U to precision ε with $O\big(\log^c \frac{1}{\varepsilon}\big)$ gates from a finite universal gate set, where c is a small constant ($c \approx 4$ in the standard proof).

11 Quantum circuits

Consider an arbitrary $f : \{0,1\}^n \to \{0,1\}^m$. It's not reversible. We want to implement

$$f^{\text{rev}} : \{0,1\}^n \times \{0,1\}^m \to \{0,1\}^n \times \{0,1\}^m, \qquad (x, y) \mapsto \big(x,\, y \oplus f(x)\big).$$

Unitary O_f: $|x\rangle\,|y\rangle \mapsto |x\rangle\,|y \oplus f(x)\rangle$.
The circuit size of O_f is the same as f:

1. Replace any copy with CNOT using scratch bits
2. Replace gates with reversible versions
3. Copy and uncompute to clean trash

UNIVERSAL GATES $\to \{\text{CNOT}\} \cup U(2)$, $\{H, \text{Toffoli}\}$, $\{\text{CNOT}, H, T\}$,
but not $\{\text{CNOT}, H, S = T^2\}$.

Most n-qubit unitaries require exponential circuit size, $t \sim \Omega(2^n)$.

BIT ORACLE: $O_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$
PHASE ORACLE: $U_f\,|x\rangle = (-1)^{f(x)}\,|x\rangle$

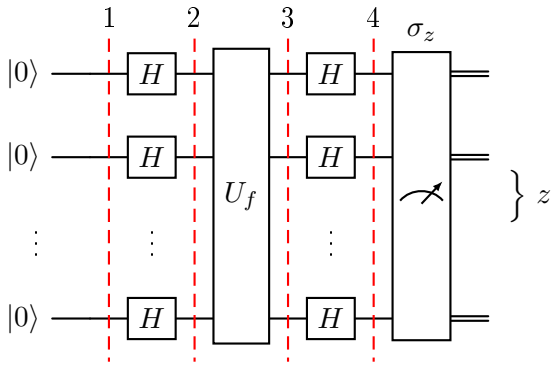
We assume to have oracles available which we can query. The number of queries an algorithm needs is called QUERY COMPLEXITY.

($\to$ Not possible to write Toffoli with only CNOT and H, 'cause, since $\{H, \text{Toffoli}\}$ is universal, that would imply $\{H, \text{CNOT}\}$ universal. But H and CNOT are Clifford gates, which can be simulated efficiently classically (Gottesman–Knill).)

12 Deutsch–Jozsa algorithm

Given a computable function $f : \{0,1\}^n \to \{0,1\}$ with the promise that f is either constant or balanced $\big(|\{x \mid f(x) = 0\}| = |\{x \mid f(x) = 1\}| = 2^{n-1}\big)$, we want to determine if f is constant or balanced with as few queries as possible.

CLASSICAL: worst case requires $2^{n-1} + 1$ calls.
QUANTUM: just 1 call to U_f.



$$|\psi_1\rangle = |0\rangle^{\otimes n}$$

$$|\psi_2\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} |x\rangle \qquad\qquad H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y} |y\rangle$$

$$|\psi_4\rangle = H^{\otimes n} \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} |x\rangle = \frac{1}{2^n} \sum_{x,y\in\{0,1\}^n} (-1)^{f(x)} (-1)^{x\cdot y} |y\rangle$$

$$\langle 0^{\otimes n}|\psi_4\rangle = \frac{1}{2^n} \sum_{x,y\in\{0,1\}^n} (-1)^{f(x)} (-1)^{x\cdot y} \overbrace{\langle 0^{\otimes n}|y\rangle}^{\delta_{0y}}$$

$$= \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{f(x)} = \begin{cases} \pm 1 & f \text{ constant} \\ 0 & f \text{ balanced} \end{cases}$$

$$\Pr[z = 0\cdots 0] = \begin{cases} 1 & f \text{ constant} \\ 0 & f \text{ balanced} \end{cases} \qquad \checkmark$$

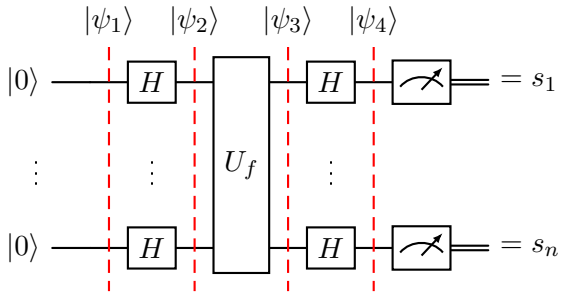
13 Bernstein–Vazirani algorithm

It's a special version of Deutsch–Jozsa. Given $f : \{0,1\}^n \to \{0,1\}$, for a given $s \in \{0,1\}^n$,

$$f(x) = x \cdot s \bmod 2 = x_1 s_1 + \cdots + x_n s_n \bmod 2,$$

find s.

- Classical: n calls to f
- Quantum: only 1 call



As in DJ,

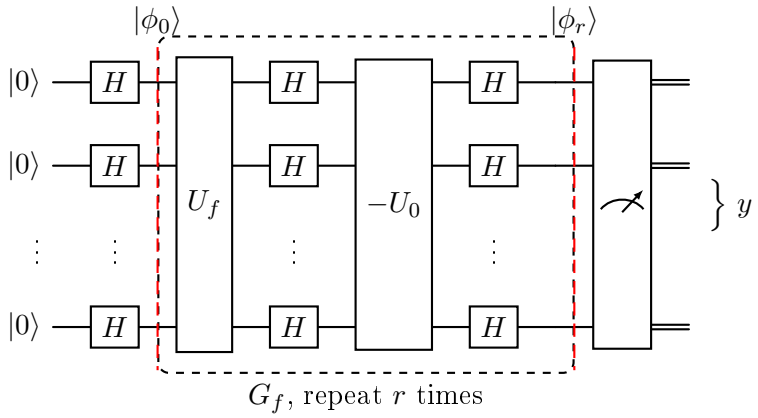
$$|\psi_4\rangle = \frac{1}{2^n} \sum_{x,y\in\{0,1\}^n} \overbrace{(-1)^{f(x)}}^{(-1)^{x\cdot s}} (-1)^{x\cdot y} |y\rangle \;\Rightarrow\; |\psi_4\rangle = \sum_y \underbrace{\left(\frac{1}{2^n} \sum_x (-1)^{x\cdot(y\oplus s)}\right)}_{= \begin{cases} 1 & y = s \\ 0 & y \neq s \end{cases}} |y\rangle = |s\rangle$$

We measure the state $|s\rangle = |s_1 \cdots s_n\rangle$ with 100% probability.

14 Grover's algorithm

Unstructured search problem: $f : \{0, \dots, N-1\} \to \{0, 1\}$ with the promise that $f(x_0) = 1$ for a single $x_0 \in \{0, \dots, N-1\}$ (MARKED ELEMENT).
$\to$ Find the marked element x_0 with as few queries as possible.

- CLASSICAL: $\Theta(N)$ queries
- QUANTUM: $r = \Theta(\sqrt{N})$, with probability $1 - O(\frac{1}{N})$



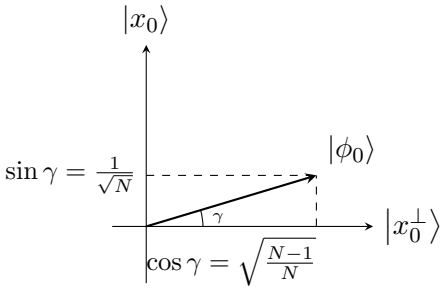
$$U_f |x\rangle = \begin{cases} -|x_0\rangle & x = x_0 \\ |x\rangle & \text{else} \end{cases}$$

$$U_f = -R_{x_0} = I_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$$
$$H^{\otimes n} U_0 H^{\otimes n} = \mathbb{I} - 2|+\rangle\langle +| = I_{\phi_0}$$

$$\Rightarrow \boxed{G_f = -R_{\phi_0} \cdot R_{x_0}} \qquad \text{Grover operator}$$

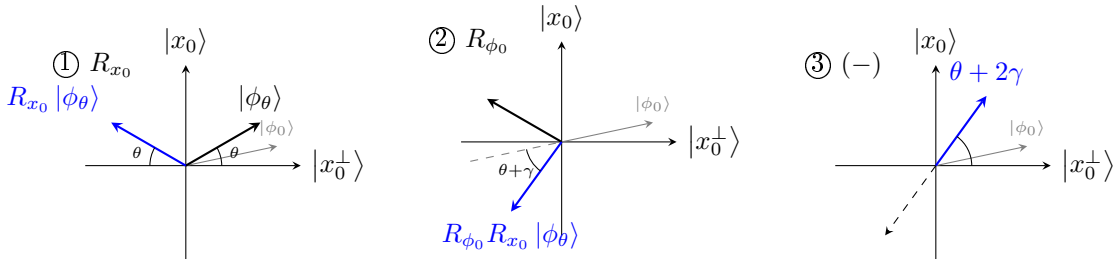
We define $\left|x_0^\perp\right\rangle := \frac{1}{\sqrt{N-1}} \sum_{x \neq x_0} |x\rangle$ ("bad state"). Then

$$|\phi_0\rangle = \frac{1}{\sqrt{N}} |x_0\rangle + \sqrt{\frac{N-1}{N}} \left|x_0^\perp\right\rangle = \sin\gamma\, |x_0\rangle + \cos\gamma \left|x_0^\perp\right\rangle .$$



Since the Grover operator is composed of rotations around $|x_0\rangle$ and $|\phi_0\rangle$, and both are in the plane spanned by $\{|x_0\rangle, \left|x_0^\perp\right\rangle\}$, we can analyze geometrically in this plane.

$W := \text{span}\{|x_0\rangle, \left|x_0^\perp\right\rangle\} \;\to\; G_f : W \to W$.
If the state is $|\phi_\theta\rangle = \sin\theta\, |x_0\rangle + \cos\theta \left|x_0^\perp\right\rangle$, G_f acts on the angle θ s.t. $\theta \mapsto 2\gamma + \theta$.



For $k \in \mathbb{N}$:

$$|\phi_k\rangle = (G_f)^k |\phi_0\rangle = \sin(\gamma + 2k\gamma)\,|x_0\rangle + \cos(\gamma + 2k\gamma)\left|x_0^\perp\right\rangle.$$

The probability to get outcome x_0 is

$$P_r = \left|\langle\phi_r|x_0\rangle\right|^2 = \left|\sin[\gamma(1+2r)]\right|^2.$$

If we want $P_r \approx 1$, then

$$\gamma(1+2r) \approx \frac{\pi}{2} + m\pi \quad \Rightarrow \quad r = \frac{2m+1}{4\gamma}\pi - \frac{1}{2} = \frac{(2m+1)\pi}{4\arcsin\left(\frac{1}{\sqrt{N}}\right)} - \frac{1}{2}.$$

If we assume $N \gg 1$, $\arcsin\left(\frac{1}{\sqrt{N}}\right) = \frac{1}{\sqrt{N}} + O\left(\frac{1}{\sqrt{N}^3}\right)$

$$\Rightarrow \boxed{r = \frac{\pi}{4}\sqrt{N} - \frac{1}{2} + \underbrace{O(1)}_{\text{round to closest integer}} \in \mathbb{N}}$$

$$P_r = \left|\sin\left(\frac{\pi}{2} + O\left(\frac{1}{\sqrt{N}}\right)\right)\right|^2 = \boxed{1 - O\left(\frac{1}{N}\right)}$$

After a rotation by π, the overlap with $-|x_0\rangle$ is high.
Period: $\Delta r = \frac{\pi}{2}\sqrt{N} + O(1)$.

It can be proven that any quantum algorithm solving the search problem has to query the oracle $\Omega(\sqrt{N})$ times
$\Rightarrow$ Grover is OPTIMAL.

14.1 Multiple marked elements

There is a set S of $M > 1$ marked elements. The circuit is the same, but

$$U_f = \mathbb{I} - 2\Pi_S, \qquad \Pi_S = \sum_{x\in S} |x\rangle\langle x|, \qquad |S\rangle = \frac{1}{\sqrt{M}}\sum_{x\in S}|x\rangle.$$

In the same way as the $M = 1$ case, we write

$$|\phi_0\rangle = \sin\gamma\,|S\rangle + \cos\gamma\left|S^\perp\right\rangle, \qquad \sin\gamma = \sqrt{\frac{M}{N}}.$$

After r iterations:

$$P_r = \left|\langle\phi_r|S\rangle\right|^2 = \sin^2\left((2r+1)\arcsin\sqrt{\frac{M}{N}}\right) \quad \text{and} \quad \boxed{r \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}} \qquad M \ll N.$$

If we don't know M, try $M' = 1$; if it fails, $M' = 2, \ldots$
The total number of queries is $O(\sqrt{N})$.

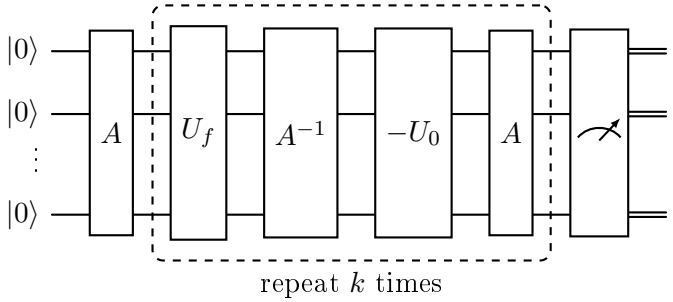
15 Amplitude amplification

Generalization of Grover's algorithm, with a set of marked elements S, $|S| = M$, and we have a heuristic guessing algorithm A:

$$A\,|0\rangle^{\otimes n} = \sum_{x\in\{0,1\}^n} \alpha_x\,|x\rangle =: |\psi\rangle\,.$$

$P = P^2$ good projector, $\langle\psi|P|\psi\rangle = g$: probability to be in the good state.

The goal is to increase the overlap with the good state, i.e. increase g.
The circuit is the same as Grover, but replacing $H^{\otimes n}$ with A:



The good state can be defined as

$$|g\rangle = \frac{P\,|\psi\rangle}{\sqrt{g}}, \qquad \left|g^\perp\right\rangle = |b\rangle = \frac{(\mathbb{I}-P)\,|\psi\rangle}{\sqrt{1-g}}.$$

Here $\sin\gamma = \sqrt{g}$, $\cos\gamma = \sqrt{1-g}$, and

$$G_f = -R_{|\psi\rangle}R_{|g\rangle},$$

where $R_{|g\rangle} = 2P - \mathbb{I}$ is the reflection around $|g\rangle$, and $R_{|\psi\rangle} = -A\,U_0A^{-1} = 2\,|\psi\rangle\langle\psi| - \mathbb{I}$ is the reflection about the state $|\psi\rangle = A\,|0\rangle^{\otimes n}$.

$\Rightarrow$ Rotation of 2γ from $|\psi\rangle$ towards $|g\rangle$ (same as Grover) in the plane $|g\rangle\,, |b\rangle$.
The state after k iterations is

$$G_f^k\,|\psi\rangle = \sin\big(\gamma(2k+1)\big)\,|g\rangle + \cos\big(\gamma(2k+1)\big)\,|b\rangle\,, \qquad \big|\langle\psi_k|g\rangle\big|^2 = \sin^2\big(\gamma(2k+1)\big).$$

For small g, $\gamma = \arcsin\sqrt{g} \approx \sqrt{g} \Rightarrow k \approx \left\lfloor \frac{\pi}{4\sqrt{g}} \right\rfloor = O\Big(\frac{1}{\sqrt{g}}\Big)$
$\rightarrow$ improvement over $O\big(\sqrt{N/M}\big)$ if $g > \frac{M}{N}$.
Overall circuit size $\approx O\big(k(s+n)\big)$, with s = circuit size of f.

- Grover is a.a. where the heuristic gives uniform probability ($A = H^{\otimes n}$).

16 Quantum Fourier transform

$\mathcal{H} = \mathbb{C}^d$ $(d = 2^n)$, $\quad \omega_d := e^{\frac{2\pi i}{d}}$.
Define

$$\mathrm{QFT}_d = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1}\sum_{k=0}^{d-1} \omega^{jk}\,|j\rangle\langle k|\,, \qquad \mathrm{QFT}_d\,|k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} e^{2\pi i\frac{k\cdot j}{d}}\,|j\rangle\,.$$

This is unitary: $\mathrm{QFT}^\dagger\mathrm{QFT} = \mathbb{I}$.

Binary notation: $y = y_0 + 2y_1 + \cdots + 2^{n-1}y_{n-1} = \sum_{k=0}^{n-1} 2^k y_k, \quad |y\rangle = |y_{n-1}\rangle\,|y_{n-2}\rangle\cdots|y_0\rangle$.
Binary fraction: $0.x_j x_{j-1}\ldots x_0 = \sum_{k\le j} \frac{x_k}{2^{j-k+1}}$.

We can derive another representation for the action of QFT on a state:

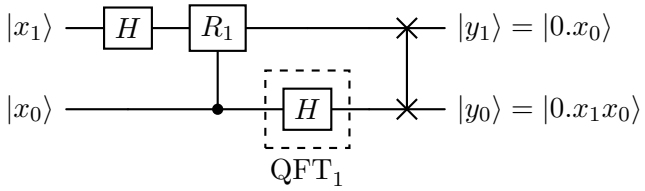
$$\begin{aligned}
\mathrm{QFT}_d\,|x\rangle &= \frac{1}{\sqrt{d}}\sum_{y=0}^{d-1} e^{\frac{2\pi i}{d}x\cdot\left(\sum_{k=0}^{n-1}2^k y_k\right)}\,|y_{n-1}\rangle\cdots|y_0\rangle \\
&= \bigotimes_{j=1}^{n}\Big(\frac{1}{\sqrt{2}}\sum_{y_{n-j}=0}^{1} e^{\frac{2\pi i}{2^j}x\,y_{n-j}}\,|y_{n-j}\rangle\Big) \\
&= \bigotimes_{j=1}^{n}\frac{1}{\sqrt{2}}\Big(|0\rangle + e^{2\pi i\,0.x_{j-1}\ldots x_0}\,|1\rangle\Big) \\
&= |0.x_0\rangle\,|0.x_1x_0\rangle\cdots|0.x_{n-1}\ldots x_0\rangle
\end{aligned}$$

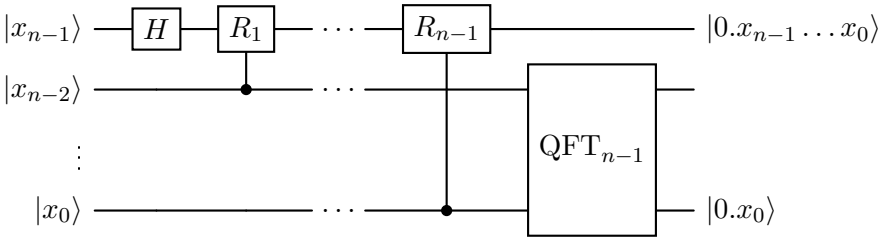
where $|0.b\rangle := \frac{1}{\sqrt{2}}\big(|0\rangle + e^{2\pi i\,0.b}\,|1\rangle\big)$. Side note: for $y_{n-j}=1$ the phase is $2\pi i\,\frac{x}{2^j} = \sum_k 2\pi i\,x_k 2^{k-j}$; the terms with $k \ge j$ are integer multiples of $2\pi i$ and drop out, the terms with $k < j$ give $0.x_{j-1}\ldots x_0$.

We can implement an efficient circuit for QFT_d ($d = 2^n$) recursively.
$n = 1 \;\to\; \mathrm{QFT}_2 = H$.

$$R_k := \mathrm{diag}\big(1, e^{\pi i/2^k}\big) = \mathrm{diag}\big(1, e^{2\pi i\,0.\overbrace{0\ldots0}^{k}x}\big)$$

We apply a sequence of controlled-R_k to the next qubit:





In the end we have to swap to get the right order: $|0.x_0\rangle\,|0.x_1x_0\rangle\cdots|0.x_{n-1}\ldots x_0\rangle$.

Total number of gates: $n + (n-1) + \cdots = \frac{n(n+1)}{2} = O(n^2)$.

- $\mathrm{QFT}\,|0\rangle^{\otimes n} = H^{\otimes n}\,|0\rangle^{\otimes n} = \frac{1}{\sqrt{d}}\sum_{k=0}^{d-1}|k\rangle$,
  but $\mathrm{QFT}\,|1\rangle\otimes|0\rangle^{\otimes n-1} = \frac{1}{\sqrt{d}}\sum_{k=0}^{d-1}(-1)^k\,|k\rangle \neq H^{\otimes n}\,|1\rangle\otimes|0\rangle^{\otimes n-1}$.
- Generalized gates:

$$Z_d = \sum_{j=0}^{d-1} e^{2\pi i\frac{j}{d}}\,|j\rangle\langle j| \qquad \to \text{eigenstates } |j\rangle,\ \lambda = e^{2\pi i\frac{j}{d}}$$

$$X_d = \sum_{j=0}^{d-1} |j\oplus 1\rangle\langle j| \qquad \to \text{eigenstates } \mathrm{QFT}_d\,|j\rangle,\ \lambda = e^{-\frac{2\pi i}{d}j}$$

$\Rightarrow$ Generalized $H_d\,|x\rangle = \frac{1}{\sqrt{d}}\sum_{y=0}^{d-1} e^{2\pi i\frac{xy}{d}}\,|y\rangle$, because it takes eigenvectors of Z_d to eigenvectors of X_d.

$$(\mathbb{I}\otimes \mathrm{QFT}_d^\dagger)\,CZ_d\,(\mathbb{I}\otimes \mathrm{QFT}_d)\,|x\rangle\,|j\rangle = |x\rangle\,|j+x \pmod d\rangle$$

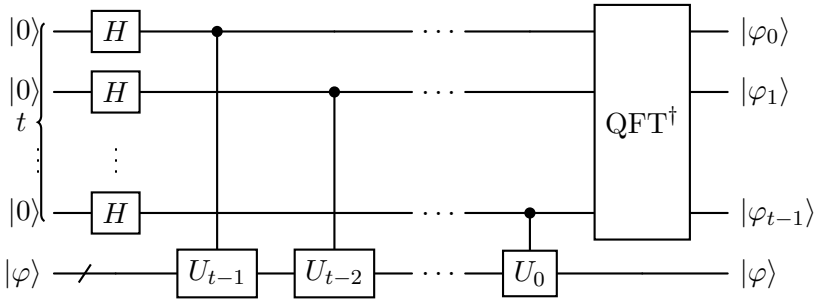
(if $d=2$: $H_2\,CZ\,H_2 = \mathrm{CNOT}$)

- $\boxed{\dfrac{1}{d}\displaystyle\sum_{j=0}^{d-1} e^{-\frac{2\pi i}{d}jk} = \delta_{0,k\,(\mathrm{mod}\ d)}}$ $\qquad \sum_{x=0}^{2^n-1}(-1)^{x\cdot y} \neq 0 \iff y=0$

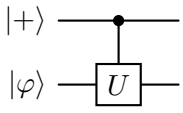
17 Quantum phase estimation

Given a unitary U and $|\varphi\rangle$: $U\,|\varphi\rangle = e^{2\pi i\varphi}\,|\varphi\rangle$, $\varphi\in[0,1)$, the goal is to find φ to t-bit precision:

$$\varphi = 0.\varphi_{t-1}\varphi_{t-2}\ldots\varphi_0 = \sum_{j=1}^{t}\varphi_{t-j}2^{-j}.$$



We apply controlled $U_k = U^{2^k}$ to create the state $|0.\varphi_0\rangle\,|0.\varphi_1\varphi_0\rangle\cdots|0.\varphi_{t-1}\ldots\varphi_0\rangle$ and then apply the inverse QFT $\to |\varphi_0\rangle\,|\varphi_1\rangle\cdots|\varphi_{t-1}\rangle$.



$$\begin{aligned}|+\rangle \to \tfrac{1}{\sqrt{2}}\big(|0\rangle\,|\varphi\rangle + U\,|1\rangle\,|\varphi\rangle\big) &= \tfrac{1}{\sqrt{2}}\big(|0\rangle + e^{2\pi i\varphi}\,|1\rangle\big)\,|\varphi\rangle\\ &= R_z(\varphi)\,|+\rangle = |0.\varphi_{t-1}\ldots\varphi_0\rangle\end{aligned}$$

$\Rightarrow R_z(2^k\varphi) = U^{2^k}$ and $R_z(2^k\varphi)\,|+\rangle = |0.\varphi_{t-k-1}\ldots\varphi_0\rangle$.

For an arbitrary $\varphi\in[0,1)$, if we want n-bit precision with probability $1-\varepsilon$:

$$t = n + \left\lceil \log\!\left(2+\frac{1}{2\varepsilon}\right)\right\rceil.$$

In the general case, $O(2^t)$ gates are required for implementing QPE.

18 Order finding

For a coprime ($\gcd(N,a)=1$) to $N\le 2^n$, consider the unitary

$$U\,|x\rangle = \begin{cases}|a\cdot x \pmod N\rangle & x\le N-1\\ |x\rangle & \text{else}\end{cases}$$

It is unitary, because $U_a^r\,|x\rangle = \Big|\overbrace{a^r}^{=1\,(\mathrm{mod}\,N)}\ x \bmod N\Big\rangle = |x\rangle$.

The goal is to find the ORDER r: $a^r \bmod N = 1$ (smallest r).

Eigenvectors of U are the states

$$\left|\varphi^{(s)}\right\rangle = \frac{1}{\sqrt{r}} \sum_{k\in\mathbb{Z}_r} e^{-2\pi i \frac{sk}{r}} \left|a^k \pmod N\right\rangle \qquad \text{with eigenvalues } \lambda_s = e^{+2\pi i \frac{s}{r}}.$$

We use QPE to find λ_s (that is $\frac{s}{r}$).
We don't know $\left|\varphi^{(s)}\right\rangle$, but $|1\rangle$ is an equal superposition of all $\left|\varphi^{(s)}\right\rangle$, $s \in \mathbb{Z}_r$:

$$|1\rangle = \frac{1}{\sqrt{r}} \sum_{s\in\mathbb{Z}_r} \left|\varphi^{(s)}\right\rangle.$$

Let $\left(\frac{s}{r}\right) = \sum_{k<t} \left(\frac{s}{r}\right)_k 2^{-k}$ (binary rep.). At the end of QPE we have the state

$$\sum_{s<r} \frac{1}{\sqrt{r}} \left|\left(\tfrac{s}{r}\right)_0\right\rangle \left|\left(\tfrac{s}{r}\right)_1\right\rangle \cdots \left|\left(\tfrac{s}{r}\right)_{t-1}\right\rangle \left|\varphi^{(s)}\right\rangle.$$

If we measure the first t qubits we collapse the superposition to a random s with equal probability $1/r$.
$\rightarrow$ We get an approximation up to t bits of precision of s/r!

Then we use the classical CONTINUED FRACTION ALGORITHM: if the output φ' satisfies $\left|\varphi' - \frac{s}{r}\right| \leq \frac{1}{2r^2}$, CF returns $\frac{s}{r}$ in lowest terms; running QPE twice (independent outputs φ, φ') and taking the l.c.m. of the two denominators gives r with probability $\geq \frac{1}{4} \rightarrow O(n^3)$.
To satisfy the bit precision, $t = 2n + 1 + \left\lceil \log\left(2 + \frac{1}{2\varepsilon}\right) \right\rceil$.

The last problem is how to build 2^{t-1} C-U gates. Fortunately, there is a smart way:

$$U_a^{2^k} : x \mapsto a^{2^k} x \pmod N.$$

We compute $z_k = a^{2^k} \bmod N = z_{k-1}^2 \bmod N$ recursively with Boolean circuits, then implement $(y, x) \mapsto (y, z_k^y x \bmod N)$ as a controlled quantum gate $\rightarrow O(t \cdot n^2)$.

$\Rightarrow$ Overall complexity is $\boxed{O(n^3)}$!

19 Shor's algorithm

The task is, given a number N (n bit), find a factor $q < N$.
Best known classical algorithm (number field sieve) $\rightarrow e^{O(n^{1/3}(\log n)^{2/3})}$, superpolynomial in n.

1. Pick a random a: $\gcd(a, N) = 1$ (if we see gcd $\neq 1$, it's a factor. We can check efficiently with Euclid's algorithm).

2. Find the order of a, r, defined as the smallest that satisfies $a^r = 1 \pmod N$ (QPE order finding, $O(n^3)$).
   With high probability ($\geq \frac{1}{2}$), r is even and neither $(a^{r/2} + 1)$ nor $(a^{r/2} - 1)$ is a multiple of N:

$$a^r - 1 = 0 \pmod N = \underbrace{(a^{r/2} + 1)}_{\neq 0 \bmod N \text{ with } p \geq \frac{1}{2}} \overbrace{(a^{r/2} - 1)}^{\neq 0 \bmod N \text{ 'cause } r \text{ is the smallest with this property}}$$

   $\Rightarrow (a^{r/2} + 1)$ must share factors with N.

3. Compute $\gcd\big((a^{r/2} + 1), N\big) = p$, then $q = \frac{N}{p}$ is the output (or simply p).

$\rightarrow$ Including all the checks during the algorithm, $O(n^3)$.

20 Density operators

Consider an ensemble $\{(p_x, |\varphi_x\rangle)\} \to$ density operator $\rho := \sum_x p_x |\varphi_x\rangle\langle\varphi_x|$.

- Probability given x is the Born rule $\langle\varphi_x|\Pi_y|\varphi_x\rangle$. Average over all x:

$$\Pr[\Pi_y] = \sum_x p_x \langle\varphi_x|\Pi_y|\varphi_x\rangle = \mathrm{Tr}\Big[\Pi_y\Big(\sum_x p_x |\varphi_x\rangle\langle\varphi_x|\Big)\Big] = \mathrm{Tr}(\Pi_y\rho)$$

- $\mathrm{Tr}[\rho] = 1$
- $\rho \geq 0 \iff \forall |\psi\rangle \in \mathcal{H}$, $\langle\psi|\rho|\psi\rangle \geq 0$ (that is, positive semi-definite), and $\rho = \rho^\dagger$

The ensemble is NOT unique ($\{(\frac{1}{2}, |0\rangle), (\frac{1}{2}, |1\rangle)\}$ and $\{(\frac{1}{2}, |+\rangle), (\frac{1}{2}, |-\rangle)\}$ both have $\rho = \mathbb{I}/2$).

20.1 Marginal states

Bipartite state $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$. We can get the marginal state ρ_A by tracing out B. We call Tr_B the PARTIAL TRACE:

$$\rho_A = \mathrm{Tr}_B\big(|\psi\rangle\langle\psi|_{AB}\big) \;\to\; \text{always a valid state.}$$

If $\Pi_y = \Pi_A \otimes \mathbb{I}_B$: $\quad \Pr(\Pi_y) = \mathrm{Tr}_A\big[\Pi_A \,\mathrm{Tr}_B\big(|\psi\rangle\langle\psi|_{AB}\big)\big]$.

Entangled states have mixed marginal states.
We call PURIFICATION of ρ_A a $|\psi\rangle_{AB}$, for some $\mathcal{H}_B$, s.t. $\rho_A = \mathrm{Tr}_B\big(|\psi\rangle\langle\psi|_{AB}\big)$.

21 Quantum channels

Closed system: $|\psi\rangle \mapsto U|\psi\rangle$.
Open system: interacts with the environment, so we apply closed-system dynamics to system + env.:

$$|\psi\rangle_A \otimes |0\rangle_E \mapsto U_{AE}\big(|\psi\rangle_A \otimes |0\rangle_E\big).$$

To find a description of the dynamics of A, we apply U_{AE} and trace out E:

$$\begin{aligned}\rho_A &= \mathrm{Tr}_E\big[U_{AE}\big(|\psi\rangle\langle\psi|_A \otimes |0\rangle\langle 0|_E\big)U_{AE}^\dagger\big] \\ &= \sum_{j=0}^{d_E-1} K_A(j)\,|\psi\rangle\langle\psi|_A\,K_A(j)^\dagger \\ &= \mathcal{N}\big[|\psi\rangle\langle\psi|\big] \qquad \text{QUANTUM CHANNEL}\end{aligned}$$

$\mathcal{N}$ is represented by KRAUS OPERATORS $\{K_A(j)\}_j$.

- $\sum_j K_j^\dagger K_j = \mathbb{I} \iff \mathrm{Tr}\big[\mathcal{N}[M]\big] = \mathrm{Tr}(M) \quad \to$ TRACE PRESERVING
- $\mathcal{N}_A \otimes \mathbb{I}_B[\rho_{AB}] \geq 0 \quad \to$ COMPLETELY POSITIVE

$$K(j) := {}_E\langle j|\,U_{AE}\,|0\rangle_E$$

$\to$ Pauli channel: $\rho \mapsto \sum_{j=0}^3 p_j \sigma_j \rho \sigma_j^\dagger, \quad K(j) = \sqrt{p_j}\,\sigma_j$

$\to$ Bit flip: $\rho \mapsto (1-p)\rho + p\,\sigma_x\rho\sigma_x; \quad K(0) = \sqrt{1-p}\,\mathbb{I}$, $K(1) = \sqrt{p}\,\sigma_x$

→ Dephasing: $\rho \mapsto (1-p)\rho + p\,\sigma_z\rho\sigma_z$; $\quad K(0) = \sqrt{1-p}\,\mathbb{I}$, $K(1) = \sqrt{p}\,\sigma_z$

→ Amplitude damping: with prob. γ, $|1\rangle \to |0\rangle$:

$$K(0) = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|\,, \qquad K(1) = \sqrt{\gamma}\,|0\rangle\langle 1|$$

Measurement is a channel too: $K(j) = \Pi_j \otimes |j\rangle$.

$$\rho_a \mapsto \rho_{ax} = \sum_j \underbrace{\left(\Pi_j \rho_a \Pi_j\right)}_{\text{post-measurement state}} \otimes \overbrace{\left(|j\rangle\langle j|\right)_x}^{\text{outcome}}$$

$$\rho_a = \mathrm{Tr}_x[\rho_{ax}] = \sum_j \Pi_j \rho_a \Pi_j \qquad \text{non-selective measurement}$$

$$\rho_x = \mathrm{Tr}_a[\rho_{ax}] = \sum_j \underbrace{\mathrm{Tr}[\Pi_j \rho_a]}_{=p_j} |j\rangle\langle j| \qquad \to \text{probability distribution of outcomes}$$

This works for arbitrary Kraus operators $\{K_j\}$: $\Lambda_j = K_j^\dagger K_j$ generalizes projective measurement.

22 Repetition code

In the classical case:

$$\text{encode} \begin{matrix} 0 \to 000 \\ 1 \to 111 \end{matrix} \overset{\text{noise}}{\rightsquigarrow} \underset{\text{(majority vote)}}{\text{decode}} \begin{matrix} 011 \to 1 \\ 010 \to 0 \end{matrix}$$

If an error occurs with probability p, and we can correct single errors, the fail probability is $p^2(3-2p)$, and if $p < \frac{1}{2}$, there's improvement.

In the quantum case, if we want to correct BIT FLIPS:

enc: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \to \alpha\,|000\rangle + \beta\,|111\rangle$, $\quad |0\rangle_L = |000\rangle$, $|1\rangle_L = |111\rangle$.
Logical operators: $X_L = XXX$, $Z_L = ZZZ$; $\quad Z_L\,|1\rangle_L = -\,|1\rangle_L$, $X_L\,|0\rangle_L = |1\rangle_L$.

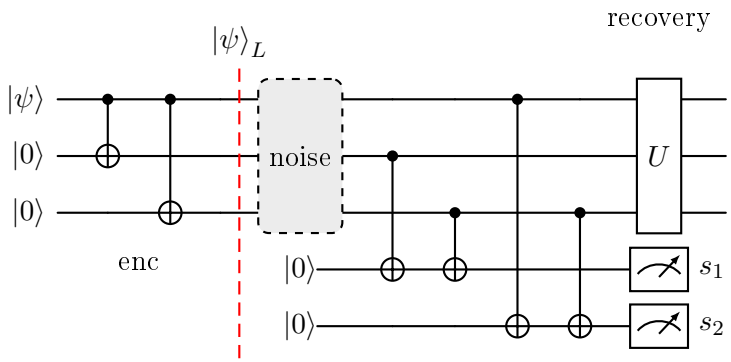
noise: $\langle X\mathbb{I}\mathbb{I},\ \mathbb{I}X\mathbb{I},\ \mathbb{I}\mathbb{I}X\rangle \to$ single-qubit bit flip

dec: SYNDROME measurements of the observables

$$S_1 = \mathbb{I}ZZ = P_0 - P_1, \qquad S_2 = Z\mathbb{I}Z,$$

where $P_0 = \mathbb{I} \otimes \left(|00\rangle\langle 00| + |11\rangle\langle 11|\right)$, $P_1 = \mathbb{I} \otimes \left(|01\rangle\langle 01| + |10\rangle\langle 10|\right)$.

S_1 and S_2 have to commute with X_L, Z_L, otherwise we would get information about the state and collapse it.
We get measurement outcomes $(-1)^{s_1}, (-1)^{s_2} \rightsquigarrow$ based on this result we apply a correction.



Ancilla 1 measures $S_1 = \mathbb{I}ZZ$ (CNOTs from qubits 2,3), ancilla 2 measures $S_2 = Z\mathbb{I}Z$ (CNOTs from qubits 1,3); a classical function $f(s_1, s_2)$ of the outcomes selects the recovery U.

Syndrome	U
$(1,1)$	$\mathbb{III}$
$(1,-1)$	$X\mathbb{II}$
$(-1,1)$	$\mathbb{I}X\mathbb{I}$
$(-1,-1)$	$\mathbb{II}X$

It fails if 2 or more errors occurred.

It can correct any single-qubit Kraus operator $K = c_0\mathbb{I} + c_1 X$, $c_0, c_1 \in \mathbb{C}$.

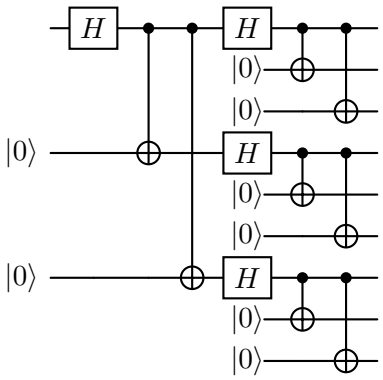
If we want to protect against PHASE FLIP errors, we use H to map $X \leftrightarrow Z$:

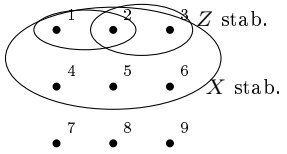
$$|+\rangle \to |+++\rangle, \qquad |-\rangle \to |---\rangle, \qquad X_L = ZZZ, \quad Z_L = XXX \text{ (equivalently } X\mathbb{II})$$

and measure $S_1 = \mathbb{I}XX$, $S_2 = X\mathbb{I}X$.
Same circuit, but with H gates on the data qubits after encoding and before decoding.

23 Shor's 9-qubit code

The idea is to concatenate encoders.





The first H makes the encoder map $|\pm\rangle \mapsto |\mp\rangle$ (below); without it one gets the usual convention $|0\rangle_L = |\hat{+}\rangle^{\otimes 3}$, $|1\rangle_L = |\hat{-}\rangle^{\otimes 3}$, with the roles of $\bar{X}$ and $\bar{Z}$ exchanged.

$$|+\rangle \to |+++\rangle \to \big(|000\rangle + |111\rangle\big)^{\otimes 3} = |\bar{+}\rangle, \qquad \big(|000\rangle + |111\rangle\big) = |\hat{+}\rangle$$
$$|-\rangle \to |---\rangle \to \big(|000\rangle - |111\rangle\big)^{\otimes 3} = |\bar{-}\rangle, \qquad \big(|000\rangle - |111\rangle\big) = |\hat{-}\rangle$$
$$\bar{X} = X_1X_2X_3, \qquad \bar{Z} = Z_1Z_4Z_7$$

Syndrome decoding:

- Measure syndromes for X errors in each block separately: $\underbrace{Z_2Z_3,\ Z_1Z_3}_{123}$, $\underbrace{Z_5Z_6,\ Z_4Z_6}_{456}$, Z_8Z_9, Z_7Z_9
- Measure syndromes for Z errors: $X_4X_5X_6X_7X_8X_9$, $X_1X_2X_3X_7X_8X_9$

$\to$ apply corrections.

- All syndrome measurement operators commute, and also with the logical operators.
- By degeneracy, Z-type errors happen only on qubits 1, 4 or 7.

- Shor's code corrects every single-qubit error $K = c_0\mathbb{I} + c_1 X + c_2 Y + c_3 Z$, $c_i \in \mathbb{C}$.
- Corrections to be applied are:
  - $\rightarrow$ for X errors, X_i, where i is the qubit with the error;
  - $\rightarrow$ for Z errors, $(ZZZ)_{\text{block } j}$, where j is the block with the error.
- Z errors on the same block are indistinguishable, e.g. $\left\langle 0_L \middle| Z_1^\dagger Z_2 \middle| 0_L \right\rangle = \left\langle 1_L \middle| Z_1^\dagger Z_2 \middle| 1_L \right\rangle \neq 0$ $\rightarrow$ indistinguishable.
  We can find a new basis of errors by diagonalizing the KL matrix C_{ab} given by $\left\langle i_L \middle| E_a^\dagger E_b \middle| j_L \right\rangle = \delta_{ij} C_{ab}$.

24 Quantum error correction

Given $\mathcal{H}_{\text{logical}} \simeq (\mathbb{C}^2)^{\otimes k}$ and noise channel $\mathcal{N}(\rho) = \sum_j E_j \rho E_j^\dagger$, we want to find

$$\text{dec} : \mathcal{H}_{\text{phys}} \to \mathcal{H}_{\log}, \qquad \text{enc} : \mathcal{H}_{\log} \to \mathcal{H}_{\text{phys}} \qquad \text{s.t.} \quad \text{dec} \circ \mathcal{N} \circ \text{enc} = \mathbb{I},$$

for some $\mathcal{H}_{\text{phys}} \simeq (\mathbb{C}^2)^{\otimes n}$.

Consider a subgroup of the n-qubit Pauli group $S = \langle s_1, \ldots, s_{n-k} \rangle \subseteq \mathcal{P}_n$. We call this the STABILIZER GROUP and define the codespace as

$$\mathcal{H}_{\text{code}} = \left\{ |\psi\rangle \in \mathcal{H}_{\text{phys}} : s\,|\psi\rangle = |\psi\rangle \ \ \forall s \in S \right\}.$$

If $[s_i, s_j] = 0\ \forall i, j$ and $-\mathbb{I}^{\otimes n} \notin S$: $\dim(\mathcal{H}_{\text{code}}) = 2^k$.

$\rightarrow$ Logical operators $\bar{X}, \bar{Z}$ s.t. $\{\bar{X}, \bar{Z}\} = 0$, and they act on the logical basis.

A collection of errors will be correctable if each class leads to a distinct syndrome.

24.1 Knill–Laflamme conditions

A code C, $\mathcal{H}_C$ corrects $\mathcal{N}(\rho) = \sum_{j=1}^m E_j \rho E_j^\dagger$

$$\iff \left\langle \psi \middle| E_i^\dagger E_j \middle| \psi' \right\rangle = C_{ij} \langle \psi | \psi' \rangle \qquad \forall \psi, \psi' \in \mathcal{H}_{\text{code}},\ C_{ij} \in \mathbb{C},\ i, j = 1, \ldots, m.$$

There are equivalent statements:

- $\left\langle \bar{k} \middle| E_i^\dagger E_j \middle| \bar{\ell} \right\rangle = \delta_{k\ell} C_{ij}$ for an ONB $\{|\bar{k}\rangle\}$ of $\mathcal{H}_{\text{code}}$
- $P E_i^\dagger E_j P = c_{ij} P$ for P the projector onto $\mathcal{H}_{\text{code}}$
- $\left\langle \bar{\psi} \middle| E_i^\dagger E_j \middle| \bar{\psi} \right\rangle = c_{ij}$ for all $|\bar{\psi}\rangle \in \mathcal{H}_{\text{code}}$

Correctability is a property of the set of errors: $\{E_j\}_j$ correctable $\Rightarrow \left\{E'_j = \sum_k M_{jk} E_k\right\}_j$ correctable ($M_{jk} \in \mathbb{C}$).

25 Stabilizer codes

Specify the stabilizer group $S \subseteq \mathcal{P}_n$:

$$\mathcal{H}_{\text{code}} = \big\{|\psi\rangle : s\,|\psi\rangle = |\psi\rangle \;\; \forall s \in S\big\}.$$

Syndrome decoding $\rightarrow$ measure the stabilizer generators and get syndromes $t_1, \ldots, t_m$, then apply Pauli corrections.

- Given S and $\{E_i\}_i$, KL are satisfied if $\forall i, j$ either

  a) $E_i^\dagger E_j \in S \quad \rightarrow$ degenerate

  b) $\exists s \in S : \{s, E_i^\dagger E_j\} = 0 \quad \rightarrow$ non-degenerate

Since two Pauli operators either commute or anticommute, the only case remaining is that E is not in S but commutes with every $s \in S \rightarrow$ not detectable:

$$\forall s \in S \quad sE\,|\psi\rangle = Es\,|\psi\rangle = E\,|\psi\rangle\,.$$

- Logical operators in stabilizer codes: $\bar{X}$ and $\bar{Z}$ s.t.
  - $\bar{X}^2 = \bar{Z}^2 = \mathbb{I}, \;\; \{\bar{X}, \bar{Z}\} = 0$
  - $[\bar{X}, s] = [\bar{Z}, s] = 0 \;\forall s \in S$
  - $\bar{X}, \bar{Z} \notin S$

We define the DISTANCE of a code as the number of qubits we need to change to go from a codeword to an orthogonal codeword $\rightarrow$ the code can correct no more than $\lfloor \frac{d-1}{2} \rfloor$ errors.

Examples are the Shor code, Steane code, surface code, 5-qubit code.

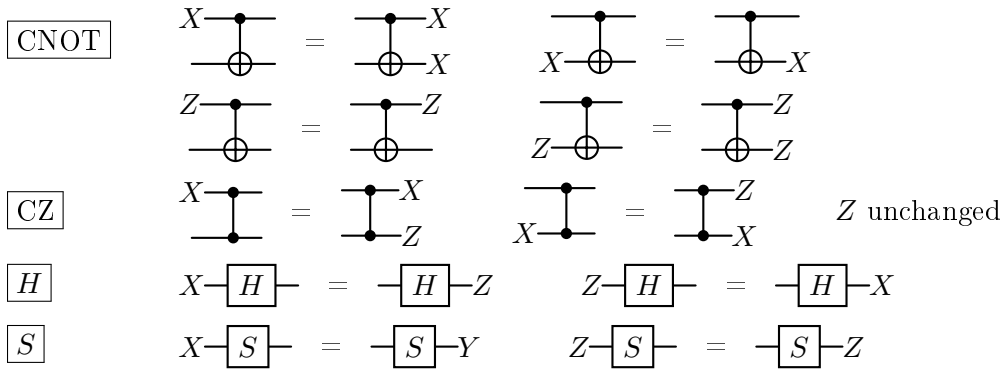
- V_S (codespace) is the intersection of the subspaces fixed by each operator s, and it's a subspace itself.
- $\text{Tr}(s_1) = \text{Tr}(i^k P_1 \otimes \cdots \otimes P_n) = i^k \prod_j \text{Tr}(P_j) = 0$, because $\text{Tr}(X) = \text{Tr}(Y) = \text{Tr}(Z) = 0$
  $\Rightarrow 2^{n-1}$ $+1$ eigenvalues, 2^{n-1} -1 eigenvalues $\rightarrow$ s_1 splits the 2^n-dim space in half.
- To find logical states, start with a "seed" that satisfies the logical Z_L condition, then apply the projector onto the $+1$ eigenstate of all stabilizers:

$$\tfrac{1}{2}(\mathbb{I} + s_1) \cdots \tfrac{1}{2}(\mathbb{I} + s_{n-k}) \underbrace{|00\ldots0\rangle}_{\text{seed}} \quad \text{(then normalize)}$$

26 Stabilizer formalism

To avoid calculations in Dirac/matrix notation, we specify the state via its stabilizers, and update them after gates or measurements.

GATE OPERATION U: if $|\psi\rangle$ is stabilized by S_k, then $U\,|\psi\rangle$ is stabilized by $US_kU^\dagger$
$\rightarrow$ from this we get gate propagation rules ($SU = U(U^\dagger SU)$).



To analyze a circuit, we start with:

- State stabilizers, if specified, e.g. $|+\rangle \to X$; $|0\rangle \to Z$
- Track (X, Z) if the state is unspecified

Then we apply the gate rules.

CLIFFORD GATES: $\mathcal{C}_n := \left\{U : UgU^\dagger \in \mathcal{P}_n \ \forall g \in \mathcal{P}_n\right\}$, generated by (H, S, CNOT).
If we track n stabilizers, each specified by $2n+2$ bits $\to O(n^2)$ memory used.

Gottesman–Knill theorem: a quantum circuit performing only Clifford gates and measurements in the computational basis can be implemented efficiently on a classical computer.

MEASUREMENT OPERATION: Pauli measurement with the stabilizer formalism.

1. Check if M commutes with all stabilizers S_i.
2. - If $[M, S_i] = 0 \ \forall i$, the measurement outcome is deterministic and the stabilizer list doesn't change.
   - If $\{M, S_i\} = 0$ for at least one i: pick one to be sacrificed, multiply it to every other anticommuting stabilizer, and replace it with $(-1)^a M$, depending on the outcome a.

27 Fault-tolerant quantum computation

An operation is defined to be fault-tolerant if a single failure in any of its physical components introduces at most one error per encoded logical block in the output → FAULT-TOLERANT GADGET.

The simplest way to construct a FT gate is to implement it TRANSVERSALLY, that is, using only independent operations on each of the physical qubits:

→ H for the Steane code: $H^{\otimes 7}$

→ CNOT for the Steane code (bitwise)

Errors can spread to the other block, but with the same weight.

Eastin–Knill theorem: no QEC code can have a universal set of gates with transversal implementation
→ other options are magic states, code switching, lattice surgery, ...

SYNDROME MEASUREMENT → a single error on an ancilla can cause multiple errors in the data
⇒ encode the ancilla and perform the measurement transversally.

28 Threshold theorem

Assuming:

1. Faults occur independently with probability p
2. QEC can correct 1 error
3. EC is ideal if: no internal faults and at most 1 input error
4. EC outputs at most 1 error if: only 1 internal fault and no input errors

The scheme fails if 2 or more errors occur, which is $c \cdot p^2 = O(G^2 p^2)$, where G is the number of gates per gadget.
There is an improvement if $cp^2 < p$, i.e. $p < p_{\text{th}} = \frac{1}{c} \simeq \frac{1}{G^2}$.

If we are in the condition $p < p_{\text{th}}$ we can improve the logical error rate by CONCATENATION: think of the FT circuit as a new circuit and implement it fault-tolerantly:

$$p \mapsto cp^2 \mapsto c(cp^2)^2 = c^3 p^4 \mapsto \cdots \mapsto c^{2^k-1} p^{2^k} = p_{\text{th}} \left(\frac{p}{p_{\text{th}}}\right)^{2^k} \xrightarrow{k\to\infty} 0$$

If above threshold, the error explodes.

- Qubit number scales as n^k.

29 FT error correction by teleportation

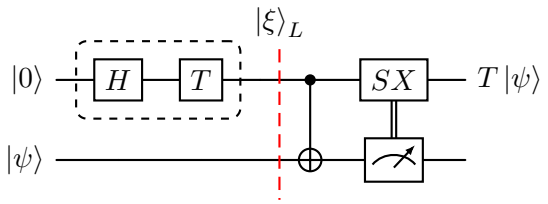
Perform a destructive Bell measurement transversally for an $[n, k]$ code.
The Bell measurement is performed bitwise: $X_{A,\alpha} \otimes X_{B,\alpha}$, $Z_{A,\alpha} \otimes Z_{B,\alpha}$ measured for $\alpha = 1, \ldots, n$ (can be done by CNOT, then X on control and Z on target).

$\rightarrow$ can reconstruct the encoded measurement by multiplying the right outcomes

$\rightarrow$ can extract the syndrome measurement from the same values

Then we apply the teleportation protocol: Bell measurement between A and B, then correction with logical $\bar{X}_L, \bar{Z}_L$ on C to teleport the state that was in A.

30 FT $\pi/8$ gate



$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$$

This is a FT implementation of T for the Steane code, assuming we can prepare fault-tolerantly the state $|\xi\rangle_L = T_L H_L |0\rangle_L$ and perform FT measurements (because S, X, CNOT can be implemented transversally with the Steane code).

How to prepare $|\xi\rangle_L$: since $Z_L |0\rangle_L = |0\rangle_L$, then $|\xi\rangle_L$ will be stabilized by

$$T_L H_L Z_L H_L T_L^\dagger = T_L X_L T_L^\dagger = e^{-i\frac{\pi}{4}} S_L X_L$$

$\Rightarrow$ Measure $e^{-i\frac{\pi}{4}} SX$:

- If we get $+1$, we have prepared $|\xi\rangle_L$.
- If we get -1, we are in the state $T_L H_L |1\rangle_L$:

  $$T_L H_L |1\rangle_L = T_L H_L X_L |0\rangle_L = T_L (H_L X_L H_L) H_L |0\rangle_L = T_L Z_L H_L |0\rangle_L = Z_L T_L H_L |0\rangle_L$$

  $\rightarrow$ apply Z_L to the final state.